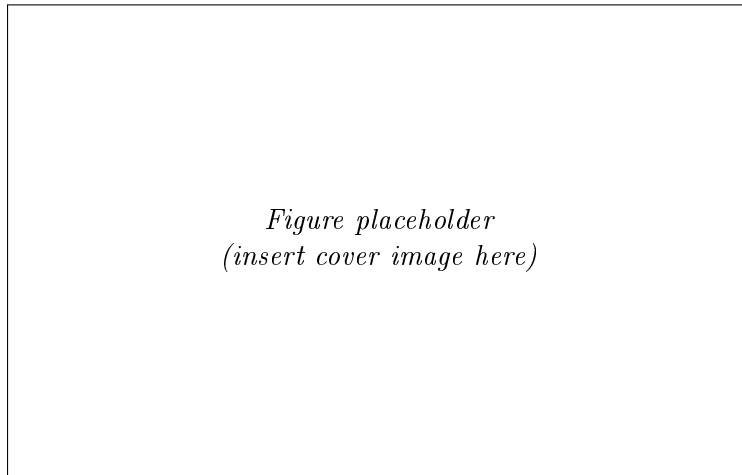


Technical Design Document

Cubli



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Space Challenges 2026

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1 Introduction

1.1 Background & Problem Statement

The Cubli is a reaction-wheel inverted pendulum: a cube-shaped platform that balances on an edge or a corner using only internal actuation, with no external supports or contact forces beyond the single line or point of contact with the ground. The concept was introduced by the Institute for Dynamic Systems and Control (IDSC) at ETH Zurich [1, 2], whose original prototype ($15 \times 15 \times 15$ cm) mounts one brushless DC motor and flywheel on each of three orthogonal faces. Commanding a change in wheel speed produces, by conservation of angular momentum, an equal and opposite reaction torque on the cube body; braking a wheel abruptly transfers its stored angular momentum to the body in a near-impulsive event, which is what allows the cube to *jump up* from resting on a face into a balance on an edge or corner.

Dynamically, the system is an *underactuated*, nonlinear, unstable-equilibrium control problem: three actuators (the wheels) must stabilize a body with more degrees of freedom than actuated inputs, about equilibria (edge and corner balance) that are open-loop unstable and increasingly coupled across axes as the contact geometry shrinks from a line (edge) to a point (corner). This is structurally the same problem as the classical inverted pendulum, extended from a single rotational axis to a fully three-dimensional, momentum-exchange-actuated body — which is why the Cubli has become a standard benchmark platform in the controls community over the past decade.

The problem this project addresses is therefore twofold: (i) derive and validate a dynamic model and controller capable of stabilizing this underactuated system about its unstable equilibria and driving it through commanded reorientations, and (ii) realize this in hardware as a modular, largely parametric design, since final actuator and sensor specifications are not yet confirmed at the time of writing. The target capability set, in order of increasing difficulty, is edge balance, corner balance, controlled rotation, and walking (Section 1.3).

Why this is a hard build, not a decorative demo. It is worth stating plainly why a machine that looks like a cube with three flywheels resists casual replication. The difficulty is not that any single subsystem is exotic — none is — but that every subsystem sits close to a physical bound, and the bounds are *coupled*: relieving one tightens another. The following failure modes recur across published and amateur replications, and each is a consequence of that coupling rather than of poor workmanship.

- **The jump-up is an energy problem with no margin.** The wheel must store, at its maximum speed, enough angular momentum to lift the cube’s center of mass over the corner. The required per-wheel inertia scales as $ML^{3/2}$ (Section 4.3.1), so mass added *anywhere* — battery, frame, fasteners — demands a larger wheel, and the wheel is itself part of M . The sizing problem is therefore a fixed-point iteration rather than a one-pass calculation, and it does not always converge: for the present cube the three-wheel set alone accounts for roughly a quarter of the total mass budget, and shrinking the wheel diameter to save space makes the situation worse, not better, because inertia must then be recovered with ballast at a smaller radius.
- **Braking torque fights the structure.** The momentum accumulated over seconds of spin-up must be shed in tens of milliseconds. That impulse passes through one motor shaft, one hub, and one motor mount — the very parts the mass budget wants thin. Many builds fail here first, and they fail in a characteristic way: the electronics are fine, the controller is sound, and the coupling, hub or mount yields on the first *successful* jump. The structural load case is set by the maneuver that proves the machine works.

- **Sensing, not the control law, sets the balancing limit.** The attitude estimate degrades from two directions at once: MEMS gyroscope bias drifts, and the accelerometer — the only absolute reference for tilt — is corrupted by vibration that the controller itself generates by spinning the wheels. Structural design and state estimation are consequently not separable concerns. A wheel half a gram out of balance at rim radius can render a perfectly correct estimator unusable, which is why wheel balance is treated here as an estimation requirement and not merely as a mechanical nicety.
- **The plant is genuinely nonlinear and constrained.** Corner balancing does not decouple into three independent single-axis problems: the axes are kinematically coupled through the contact point and the body inertia tensor. The actuators saturate in *both* torque and speed, and under any persistent disturbance the wheels drift toward their speed limit, so stored momentum must be actively managed rather than left to accumulate. Linearizing about the upright equilibrium is entirely adequate for a balancing demonstration and inadequate for anything more interesting [3].

The common thread is worth making explicit, because it determines how this project is sequenced. Replications typically fail not because the control law was wrong, but because the machine was never capable of the maneuver being asked of it: insufficient wheel inertia, motors that cannot reach their rated speed under load, a brake too slow to approximate an impulse, or a mass budget that quietly grew past the point where the stored energy suffices. Every one of these is decidable on paper, before any material is cut — and commonly is not decided there. That is the reason sizing is treated in this document as the first engineering activity rather than a detail to be settled once parts arrive.

1.2 Purpose, Application & Mission Context

The Cubli is not presented here as a balancing novelty but as a terrestrial, gravity-loaded demonstrator of **momentum-exchange attitude control** — the same physical mechanism that governs how real spacecraft point themselves. Nearly every three-axis-stabilized satellite holds and changes its attitude using reaction wheels: flywheels whose commanded speed change produces a reaction torque on the spacecraft body, exactly as in the Cubli. Spacecraft typically carry three or four such wheels (often in a pyramidal arrangement for redundancy) — directly analogous to the Cubli’s three orthogonal wheels — and this reaction-wheel approach is the standard for CubeSats and small satellites in particular, making the Cubli representative of a real, currently-flown hardware class rather than an idealized toy problem. Reaction wheels are distinct from control moment gyros (CMGs), which gimbal a constantly-spinning wheel instead of changing its speed to produce much larger torques [4]; positioning the Cubli explicitly as a *reaction-wheel*, not CMG, demonstrator keeps the application claim precise.

Framed as an ADCS problem, the Cubli’s core behaviors map directly onto a spacecraft’s day-to-day pointing task:

- **Edge / corner balance** → regulation to a commanded reference attitude and disturbance rejection — holding a pointing setpoint against perturbations.
- **Controlled rotation** → a slew maneuver: reference tracking to a new commanded attitude, exactly as a satellite re-aims a payload or antenna.
- **Jump / walk** → momentum-exchange hopping mobility, flight-proven on small bodies where wheeled traction is impossible. JAXA’s Hayabusa2 mission deployed the MINERVA-II1 rovers [5] and the MASCOT lander [6] on asteroid Ryugu (2018), both of which move by spinning up and braking an internal flywheel or eccentric mass to hop across the surface, with MASCOT additionally self-righting the same way. Walking, as a sequence of

controlled, directional hops, is a genuine step beyond a single ballistic hop and is an active research frontier (e.g. ETH's SpaceHopper [7]).

This gives the project two complementary application narratives rather than one: it is simultaneously a low-cost, hardware-in-the-loop testbed for standard spacecraft *attitude determination and control* (ADCS), and a ground analog for the momentum-exchange *surface mobility* now used on small-body missions. Framed against a companion star-tracker project, which answers “*where am I pointing?*” (attitude determination), the Cubli answers “*drive me to that reference and hold it*” (attitude control) — the two halves of a complete ADCS loop.

Why the problem is still open. The nominal Cubli problem — balance on a corner, jump up — is solved, and was solved a decade ago on well-instrumented, precisely manufactured, generously funded hardware [1, 2, 3]. Reproducing that result is an engineering exercise, not a research contribution, and this document does not claim otherwise. The interesting questions begin immediately *after* that point, and they are not Cubli-specific: they are the questions a spacecraft attitude-control engineer faces, in a form small enough to be answered on a bench.

- **Control under simultaneous torque and momentum saturation.** A jump saturates the wheels by construction — the maneuver is defined by running them to their speed limit and dumping the momentum. Steering a nonlinear attitude system to a target while respecting a finite momentum envelope, and desaturating without losing the pointing objective, is still handled largely case-by-case with tuned heuristics rather than by a general method.
- **Recovery from large attitude excursions.** Linear controllers valid near upright say nothing useful once the body is far from equilibrium, which is precisely the regime a tumbling or mis-deployed spacecraft occupies. Nonlinear and predictive methods make claims there, and verifying those claims empirically requires a system in which a failed recovery costs a re-print rather than a mission.
- **Robustness to badly known parameters.** Real inertia tensors, friction coefficients and center-of-mass offsets are known to a few percent at best, and shift with assembly, thermal state and wear. Adaptive and robust designs are overwhelmingly validated against *simulated* parameter errors, which are drawn from the same distribution the designer assumed.
- **Whether high-performance methods survive commodity sensing.** Attitude-control techniques developed against clean measurements must eventually run on a MEMS IMU with drifting bias and vibration-corrupted accelerometers. Whether a given method degrades gracefully or collapses under that sensing is an empirical question, and it is settled by hardware rather than by analysis.

Why a Cubli, and why built from scrap. The Cubli is the cheapest honest test of the four questions above. Simulation cannot answer them, because the effects that decide the outcome are exactly the ones a model omits: unmodeled friction, frame resonance excited by the wheels, and the interaction between wheel imbalance and estimator bias. These are what separate a controller that works in simulation from one that works on hardware. At the other extreme, a full-scale testbed — air bearing, vacuum chamber, flight-grade wheels — costs orders of magnitude more and, decisively, *cannot be crashed repeatedly*. The scrap-built cube occupies the useful middle: physical enough to be honest, cheap enough to be abused. The choice to build it from commodity and salvaged parts is therefore methodological rather than merely budgetary, on four counts:

- **Cheap failure means the aggressive experiments actually get run.** A controller that must never fail cannot be tested at its limits, and a testbed too valuable to break silently biases the experimental program toward maneuvers already known to work.

- **Cheap parts are a stronger test than good parts.** Loose tolerances, an imperfectly balanced wheel and a drifting IMU are not defects to be apologized for — they *are* the parameter uncertainty and disturbance content that robust and adaptive methods claim to tolerate. A demonstration on deliberately imperfect hardware therefore carries evidence that a demonstration on precision hardware does not.
- **Constraints force honest sizing.** With a fixed motor, a fixed print volume and a fixed budget, a marginal design cannot be rescued by specifying a larger component. This is the entire justification for the bolt-in-pocket ballast scheme of Section 5.3.3: it buys inertia at the outer radius using standard M6 steel bolts and no machined part, and whether it closes is a calculation that must be completed *before* printing, not discovered afterwards.
- **A commodity bill of materials is reproducible.** Every part is orderable or salvageable by any university group, so the result can be checked by others — which is the difference between a demonstration and a claim.

Mission context. The subsystem exercised here is the one a spacecraft uses for pointing: momentum-exchange attitude control under actuator saturation, with desaturation, on an underactuated body whose inertia is imperfectly known. In one respect the terrestrial case is strictly *harder* than the orbital one. In orbit, a marginal attitude controller degrades slowly and often invisibly — pointing error grows, wheels creep toward saturation, and the fault may take orbits to become obvious. Under gravity, the destabilizing torque never vanishes and never relents: an inadequate controller falls over immediately and unambiguously. The demonstration is, in that sense, self-evaluating, and this is a feature of the platform rather than a limitation of it.

1.3 Objectives

The project targets four behaviors, deliberately ordered as a difficulty ramp in which each milestone is a control precondition for the next:

1. **Edge balance** — stabilize the cube on a single edge, the least unstable equilibrium (contact is a line, one primary axis of instability).
2. **Corner balance** — stabilize the cube on a single vertex, the fully unstable equilibrium (point contact, all three axes coupled and unstable simultaneously).
3. **Controlled rotation** — execute a commanded slew between equilibria (e.g. edge to edge, or in-place reorientation), demonstrating reference tracking rather than pure regulation.
4. **Walking** — chain repeated, directional controlled falls between edges/corners into a net translation across a surface, re-stabilizing after each transition.

These map one-to-one onto the applications of Section 1.2: (1)–(2) demonstrate pointing regulation and disturbance rejection of increasing difficulty; (3) demonstrates slew/reference-tracking; (4) demonstrates momentum-exchange mobility of the kind flown on Hayabusa2. Milestones (1)–(3) are treated as committed deliverables for this design; walking (4) is scoped as the stretch objective, contingent on hardware confirmation and the jump-up feasibility check (angular momentum required to tip the cube vs. available motor/wheel torque), to be resolved once final components are known.

2 Project Organization

2.1 Team Organization

The project is developed by a multidisciplinary team, with responsibilities distributed across the main technical domains of the Cubli: mathematical modeling, control, mechanical/CAD design, and system integration. Table 1 summarizes each member's primary role, and the detailed responsibilities are outlined below.

Table 1: Team members, contact details and primary roles.

Member	Contact	Primary Role
Pablo Urioste Alarcón	pb.urioste4@gmail.com +34 654 491 174	System integration and lead: digital actuator and sensor coding, embedded software, and integration of the overall design.
Deyan Nikolaev Vlaev	vlaev.dido@gmail.com +359 877 082 499	Mathematical demonstration and derivation of the system dynamics.
Niccolo	—	Control system design and analysis.
Andrea Liang	andrea.liang.2006@gmail.com +31 06 5796 2976	Support on the control problem and structural/mechanical integration.
Suvanna Viriyanti Wu	Viriyanti.wu@gmail.com +39 329 074 7550	CAD modeling and mechanical design.
Athanasia Nikolova	—	CAD modeling, prototype building/assembly, and product presentation (commercial pitch).

Responsibilities.

- **Pablo Urioste Alarcón** — System integration and overall coordination. Handles the coding of the digital actuators and sensors, the embedded software stack, and the integration of the mechanical, electronic and control subsystems into a working prototype.
- **Deyan Nikolaev Vlaev** — Mathematical foundation. Responsible for the analytical demonstration and derivation of the equations of motion and the dynamic model of the Cubli.
- **Niccolo** — Control. Leads the design, tuning and analysis of the control strategy used to balance and maneuver the Cubli.
- **Andrea Liang** — Control and mechanics support. Assists on the control problem and contributes to the structural and mechanical integration of the system.
- **Suvanna Viriyanti Wu** — CAD and mechanical design. Develops the 3D models and mechanical drawings of the structure and reaction-wheel assembly.
- **Athanasia Nikolova** — CAD, prototype building and commercialization. Contributes to the 3D modeling, leads the physical building and assembly of the prototype, and is responsible for presenting and pitching the product.

2.2 Work Breakdown Structure

This plan runs from project start to the final presentation on 20 August 2026. The documentation stream is front-loaded and closes with the Technical Design Document deliverable on 3 August,

while the control and hardware streams continue through the build, balancing and jump-up demonstrations up to the presentation. The work is grouped into three streams — investigation, requirements and documentation; control and software; and hardware design and build — led by the owners assigned in Table 1. The investigation is kept to the essentials needed to substantiate the design: a focused literature review, the dynamics derivation, and the wheel sizing that follows from it. The table below breaks the three streams into individual tasks, each with its scope, owner and dates.

ID	Task	Scope	Owner	Dates
Investigation, Requirements & Documentation (to the 3 Aug TDD deliverable)				
A1	Literature & state of the art	Review the ETH Cubli, the one-wheel Cubli, CubeSat ADCS practice and the hopping-mission precedents	Suvanna (+Dejan)	23–25 Jul
A2	Problem statement & framing	Lock the ADCS application narrative that the rest of the document refers back to	Dejan	23–24 Jul
A3	Scope & requirements	Define scope and build the requirements table (functional, performance, design constraint) with verification methods	Dejan (+Pablo)	24–25 Jul
A4	Milestones & roles	Prototype staging, binary gate criteria, and a one-page RACI of owners	Dejan	23–25 Jul
A5	Risk register	Trigger, impact, owner and mitigation for the leading risks	All (+Pablo)	28–29 Jul
A6	1-DoF equations of motion	Lagrangian derivation of the reaction-wheel pendulum with friction terms	Dejan (+Andrea)	24–25 Jul
A7	3D equations of motion	Coupled three-wheel dynamics for the edge and corner pivots	Dejan (+Andrea)	27–28 Jul
A8	Linearisation & controllability	Equilibria, linear models, controllability/observability checks	Dejan (+Andrea)	29 Jul
A9	Jump-up sizing & budget	Energy-method wheel sizing and the parametric mass/inertia/CoM budget	Dejan (+Andrea)	30 Jul–1 Aug
A10	TDD draft (all sections)	Every member drafts their section; placeholder figures acceptable	All	25–26 Jul
A11	TDD deepen + S1 data	Full modelling and control sections; insert the rig evidence and measured parameters	All	1–2 Aug
A12	TDD assemble & submit	Consistency pass, figures, references, submission of the deliverable	All (Dejan integrates)	3 Aug
Control & Software				
B1	Sim environment + plant	Simulation setup and the 1-DoF plant, validated by energy conservation	Andrea	23–24 Jul
B2	LQR design (1-DoF, sim)	Tune the regulator and sweep robustness to inertia error, delay and saturation	Andrea (+Dejan)	26–27 Jul
B3	State estimator	Complementary tilt filter with injected noise and bias in simulation	Nicc	27–28 Jul
B4	Firmware & comms	Firmware architecture, loop rates, and the CAN-FD command/reply layer	Nicc (+Pablo)	24–28 Jul
B5	Sensor drivers + cal	IMU and encoder drivers, gyro/accel calibration and the lever-arm correction	Nicc (+Andrea)	28–30 Jul
B6	Safety & mode logic	Watchdog, tilt-limit disarm, wheel-speed cap and the mode state machine	Nicc (+Andrea)	30–31 Jul

ID	Task	Scope	Owner	Dates
B7	S1 bring-up & balance	Port estimator and LQR to the rig and tune to a stable balance	Nicc (+Andrea)	31 Jul–2 Aug
B8	System identification	Measure K_t , wheel inertia, friction and loop latency on the rig	Andrea (+Nicc)	1–2 Aug
B9	3D model + LQR (sim)	3D plant and the edge/corner regulators with the CAD inertia tensor	Andrea (+Dejan)	3–6 Aug
B10	EKF / MEKF estimator	3D estimator with gyro-bias states; characterise the yaw drift	Nicc (+Andrea)	6–8 Aug
B11	Edge balance (S2a)	Retune the edge controller on the assembled cube; log disturbance recovery	Andrea (+Nicc)	10–13 Aug
B12	Corner balance + slew (S2b/c)	Coupled three-axis corner balance and a commanded slew; report ADCS metrics	Andrea (+Nicc)	14–16 Aug
B13	Jump-up (S3)	Spin-up profile and brake release for a face-to-edge jump	Nicc (+Andrea)	17–18 Aug

Hardware Design & Build

C1	Envelope & layout study	Packaging of wheels, drivers, battery and electronics with a central CoM	Neisa (+Suvanna)	23–24 Jul
C2	Reaction-wheel design	Wheel to the inertia target: radius, rim, hub bolting to the motor bell	Suvanna (+Neisa)	24–26 Jul
C3	Mounts & brackets	Motor mount, ring-magnet carrier and adjustable encoder bracket	Neisa, Suvanna	27–29 Jul
C4	1-DoF rig CAD + jigs	Single-axis rig and the balancing/assembly tooling	Suvanna (+Neisa)	25–28 Jul
C5	Mass properties & freeze	Export the inertia tensor/CoM and release the manufacturing package	Neisa (+Suvanna)	2–3 Aug
C6	Procurement order	Order the long-lead items (rims, frame stock, connectors, spares)	Pablo (+Neisa)	23–24 Jul
C7	Bench PSU & bring-up	Current-limited supply, safety kit, and single-axis electrical bring-up	Pablo (+Nicc)	25–27 Jul
C8	Fabricate wheels ($\times 4$)	Cut/receive rims, print hubs, bond magnets — three plus one rig spare	Neisa, Suvanna	27–30 Jul
C9	Balance & spin-test	Static-balance and containment spin-test every wheel	Suvanna (+Neisa)	30–31 Jul
C10	Assemble 1-DoF rig	Motor, balanced wheel, encoder gap, IMU and hard stops	Pablo (+Neisa)	31 Jul–1 Aug
C11	Print structure + frame	Structural print set and the fabricated, orthogonal frame	Neisa, Suvanna	3–6 Aug
C12	Perfboard + harness	Power/signal board and the full power and CAN harness	Pablo	3–6 Aug
C13	Cube integration + power-on	Assemble the three axes, then bench and battery smoke test	Pablo (+Neisa)	7–9 Aug
C14	Brake mechanism build	Fabricate and fit the servo brake; bench-test the deceleration	Suvanna (+Pablo)	10–13 Aug

Electrical & Electronics (electronics design: Nicc; wiring, testing and bring-up: Pablo)

E1	Bench power, e-stop & triage	Current-limited bench supply, e-stop and inline fuse; incoming test of all three actuation sets against the 194 m Ω phase figure, with CAN IDs assigned	Pablo	28–29 Jul
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ID	Task	Scope	Owner	Dates
E2	Driver, phase wiring & 5 V rail	Motor phase wiring and FOC calibration of the first set; LM2596 rail trimmed to 5.00 V and characterised under load to 1.5 A	Pablo (+Nicc)	29–30 Jul
E3	MA600 encoder + rig CAN bus	Magnet bonding, air gap ≤ 0.5 mm and bias trim; 2-node CAN-FD bus with split termination at both physical ends	Pablo	30–31 Jul
E4	Rig parameters + board outline	Measurement setup for K_t , wheel inertia, friction and loop latency; driver placement, CAN routing and the perf-board floorplan issued as a frozen outline	Nicc (+Pablo)	31 Jul–1 Aug
E5	1-DoF closed-loop bring-up	Teensy, IMU and CAN-FD closed loop on the rig; disarm path and watchdog verified before any balance attempt	Pablo (+Nicc)	1–2 Aug
E6	Perfboard + battery power chain	Populate the distribution perfboard (two ground domains, one star point, split 5 V rail); XT90 chain with anti-spark, fuse and ≥ 50 V bulk capacitance; replicate the remaining actuation sets	Nicc (+Pablo)	3–4 Aug
E7	4-node bus + cube harness	Bus scaled to four nodes with stubs < 10 cm and an error soak at 5 Mbit/s; full power and signal harness fabricated and routed; Wi-Fi telemetry for untethered operation	Pablo	5–7 Aug
E8	Cube power-on, 48 A + thermal	First battery power-on and the three-axis 48 A transient; driver thermals in the enclosed volume; all axes addressable	Pablo (+Nicc)	7–9 Aug
E9	Brake servo rail + measurement	Servo on an isolated 5 V rail with PWM drive and flyback; rail stability under stall, brake closure time and spin-down brake torque measured	Pablo (+Nicc)	10–18 Aug
Deliverables				
D1	Results, demo & final presentation	Slow-motion capture, results plots, deck, and the final presentation	All (+Suvanna edits)	17–20 Aug

2.3 Schedule and Contingency Plan

The full-page schedule in Figure 1 places every task of Section 2.2 on the 23 July–20 August 2026 calendar, and Table 3 states the gate that closes each phase. The documentation stream is deliberately completed early, at the TDD deliverable on 3 August, so the remaining weeks are free for the build, balancing and jump-up work that culminates in the final presentation. The critical path runs through the dynamics investigation and wheel sizing, which feed both the control and the hardware streams; the single-axis (S1) rig is the protected item, since its balance result is the experimental evidence in the document. If any modeling or rig task slips, the TDD is still submitted on schedule with simulation-only evidence and the affected result carried into the later build weeks.

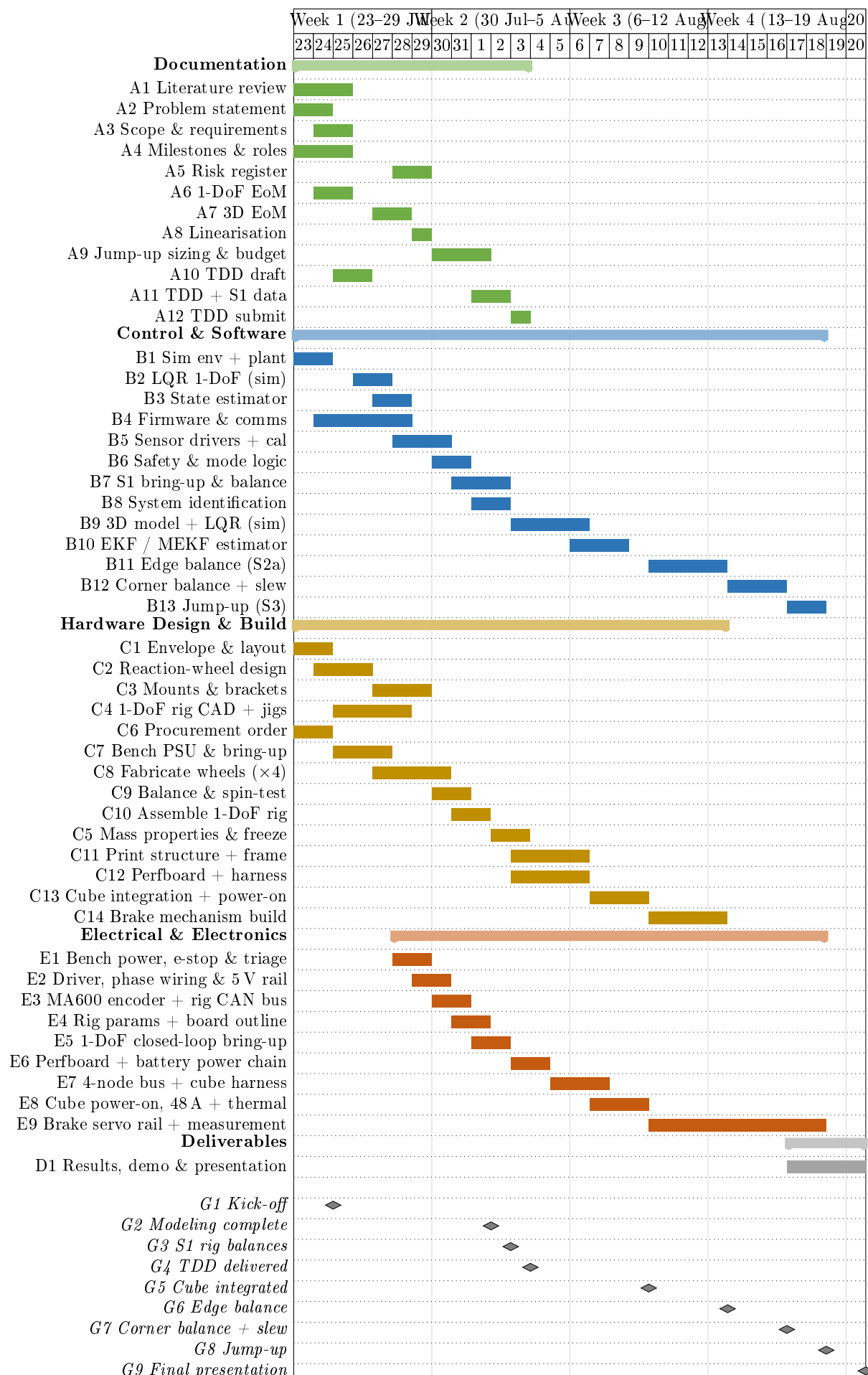


Figure 1: Detailed project schedule, 23 July–20 August 2026. The documentation stream (green) closes at the TDD deliverable on 3 August; control (blue), hardware (gold) and electrical (orange) continue through the build and demonstrations to the final presentation on 20 August. The electrical stream covers the design, wiring and testing of the 1-DoF rig (E1–E5, to G2), its extension to the 3D cube

Table 3: Milestone gates, from kick-off to the final presentation.

Gate	Pass criterion	Date
G1	Long-lead order placed; requirements and plan locked	24 Jul
G2	Dynamics, control design and wheel sizing complete	1 Aug
G3	1-DoF rig balances ≥ 60 s; plant parameters measured	2 Aug
G4	TDD delivered (documentation deliverable)	3 Aug
G5	Cube integrated and powered; all axes addressable	9 Aug
G6	Edge balance (S2a) for ≥ 60 s with disturbance recovery	13 Aug
G7	Corner balance and commanded slew (S2b/c)	16 Aug
G8	Face-to-edge jump-up (S3) captured on video	18 Aug
G9	Final presentation and demonstration	20 Aug

3 System Analysis & Requirements

3.1 Scope of Work & Use Cases

3.2 Actuation Trade-off — Why Reaction Wheels

3.3 Requirements & Objectives Traceability

Each project objective (Section 1.3) imposes a required capability, which in turn is confirmed by a specific test in the validation plan (Section 6). The mapping below keeps every requirement traceable from motivation to verification.

Table 4: Objective-to-requirement-to-verification traceability.

Objective	Required capability	Verified by
Edge balance	Regulation about a line-contact equilibrium	Sec. 6
Corner balance	Regulation about a point-contact equilibrium, all axes	Sec. 6
Controlled rotation	Reference tracking / slew between equilibria	Sec. 6
Walking (stretch)	Chained jump-up + re-stabilization	Sec. 6

3.4 Coordinate Frames & Conventions

4 Dynamic Modeling & Control

The full Cubli is underactuated and simultaneously unstable about all three axes when balanced on a corner (Section 1.3). The modeling, identification, and control workflow is therefore first validated on a two-dimensional (2D) single-face prototype with one rotational degree of freedom, before being generalized to the 3D cube in Section 4.5.

4.1 2D Reaction-Wheel Pendulum

The 2D prototype is a 3D-printed plate sized to one cube face, with a motor-driven momentum wheel at its center and a braking mechanism at one corner. The plate pivots on a bearing at that corner, constraining motion to a single rotational degree of freedom in a vertical plane.

4.1.1 Equations of Motion

The nonlinear equations of motion are

$$\ddot{\theta}_b = \frac{(m_b l_b + m_w l) g \sin \theta_b - T_m - C_b \dot{\theta}_b + C_w \dot{\theta}_w}{I_b + m_w l^2}, \quad (1)$$

$$\begin{aligned} \ddot{\theta}_w = & \frac{(I_b + I_w + m_w l^2)(T_m - C_w \dot{\theta}_w)}{I_w(I_b + m_w l^2)} \\ & - \frac{(m_b l_b + m_w l) g \sin \theta_b - C_b \dot{\theta}_b}{I_b + m_w l^2}, \end{aligned} \quad (2)$$

with motor current-torque relation $T_m = K_m u$. Symbols are defined in Table 5 and used throughout this section.

Table 5: 2D pendulum model variables.

Symbol	Description	Unit
θ_b	Body tilt angle	rad
θ_w	Wheel rotation relative to body	rad
m_b, m_w	Body, wheel mass	kg
I_b, I_w	Body (about pivot), wheel (about motor axis) inertia	kg m ²
l	Motor axis to pivot distance	m
l_b	Body CoM to pivot distance	m
g	Gravitational acceleration	m/s ²
T_m	Motor torque	N m
C_b, C_w	Body, wheel damping coefficient	kg m ² /s
K_m	Motor torque constant	N m/A
u	Motor current input	A

4.1.2 Equilibria & Linearization

The relevant equilibrium is the upright position $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$; its linearization is presented with the control design in Section 4.4.

4.2 Parameter Identification & the 2D Testbed

CAD mass-properties simulation, accounting for print material, infill, and mounted hardware (driver, encoder, MCU, gyroscope/accelerometer, wiring), gives m_b, m_w, I_b, I_w directly, while l and l_b follow from the plate's design geometry. The damping coefficients C_b and C_w cannot be obtained from CAD, since they depend on bearing friction, wiring drag, and motor cogging; nominal values will be estimated for the initial control design, with dedicated testing planned once the prototype is fabricated.

Table 6: Parameter determination method (symbols per Table 5).

Symbol	Determination method
l	Calculated during design
l_b	Calculated during design
m_b	CAD mass properties
m_w	CAD mass properties
I_b	CAD mass properties
I_w	CAD mass properties
C_b	Estimated (future testing planned with prototype)
C_w	Estimated (future testing planned with prototype)
K_m	Motor datasheet

4.3 Jump-Up Feasibility Analysis

Given CAD-derived $I_b, I_w, m_b, l_b, m_w, l$, the wheel speed ω_w required for jump-up is estimated before fabrication, assuming a perfectly inelastic wheel-body collision. Conservation of angular momentum at impact and of energy from impact to edge-standing yield

$$\omega_w^2 = (2 - \sqrt{2}) \frac{I_w + I_b + m_w l^2}{I_w^2} (m_b l_b + m_w l) g. \quad (3)$$

This value is checked against the motor's achievable speed at the available bus voltage and used to size wheel inertia (radius, rim mass) within the motor's operating range without excessively reducing the balancing recovery angle. Post-fabrication, ω_w is refined experimentally to correct for deviations from the ideal collision assumption.

The braking mechanism uses a servo-driven barrier that strikes a bolt head on the wheel to stop it abruptly. Servo response time bounds the validity of the “instantaneous stop” assumption; a slower stop yields a lower effective post-impact body velocity than predicted, compensated through the same tuning process.

4.3.1 Torque-Limited Wheel-Inertia Target

The relation above assumes an instantaneous, lossless momentum transfer. For preliminary sizing, the achievable motor speed is treated as fixed and the required per-wheel inertia is derived from a momentum budget that additionally accounts for finite brake torque. This gives an ideal per-wheel angular momentum

$$h_{w,\text{ideal}} = \sqrt{\eta} \frac{\sqrt{2g\lambda\kappa}}{n} ML^{3/2}, \quad (4)$$

where M, L, g are cube mass, edge length, and gravity, η is an energy margin, and κ, λ, n are dimensionless geometric factors set by the contact case (Table 7).

The corner case is more demanding and governs the design.

Table 7: Contact-case geometric factors.

Case	κ (pivot inertia)	λ (CoM rise)	n (momentum transfer)
Edge	2/3	$1/\sqrt{2} - 1/2$	1
Corner (governs design)	11/12	$\sqrt{3}/2 - 1/2$	$\sqrt{2}$

A finite brake torque τ_b cannot deliver the impulse instantaneously and must exceed the gravitational tip-over torque $\tau_g = MgL/2$. This loss is captured by a brake-amplification factor

$$\beta = \frac{\tau_b}{\tau_b - \tau_g}, \quad h_w = \beta h_{w,\text{ideal}}, \quad I_{w,\text{target}} = \frac{h_w}{\omega_{\max}}, \quad (5)$$

where $\omega_{\max}$ is the maximum practical wheel speed. Equations (4) and (5) are the sizing relations used at step 4 of the closure procedure of Section 5.2: given a candidate cube mass and edge length they return the inertia the wheel of Section 5.3.3 must meet. Their inputs are recorded in Table 8 and are not fixed here— M and L are outputs of the packaging closure, not assumptions preceding it.

Two features of these relations govern how the iteration behaves. First, $h_w \propto ML^{3/2}$, so growing the cube to accommodate a larger wheel also raises the requirement; convergence depends on the achievable inertia growing faster ($\sim D_w^2$) than the requirement does. Second, β diverges as $\tau_b \rightarrow \tau_g$: a brake whose torque only marginally exceeds the gravitational tip-over torque cannot deliver the required impulse at any wheel speed. The brake torque must therefore be checked against $\tau_g = MgL/2$ on every iteration, and a comfortable ratio τ_b/τ_g maintained rather than a merely positive margin.

Table 8: Torque-limited wheel-inertia target (corner case). Inputs are fixed by the closure procedure of Section 5.2; the table is populated on each iteration.

Symbol	Description	Value
M	Cube mass	TBD
L	Cube edge length	TBD
η	Energy margin	TBD
τ_b	Brake torque per wheel	TBD
$\tau_g = MgL/2$	Gravitational tip-over floor	TBD
β	Brake-amplification factor	TBD
$\omega_{\max}$	Max wheel speed (design)	TBD
h_w	Required angular momentum	TBD
$I_{w,\text{target}}$	Target wheel inertia	TBD

Choice of $\omega_{\max}$. The maximum wheel speed is a design choice rather than a datasheet figure, and it enters the sizing directly through $I_{w,\text{target}} = h_w/\omega_{\max}$: a higher assumed speed buys a proportionally smaller wheel. It is therefore the natural relief valve when the envelope caps the wheel below the requirement (step 5, Table 11), but it must be set from what the drive can sustain in service, not from the no-load figure $K_v V_{\text{bus}}$. Three effects separate the two: winding resistance, which costs speed under the torque required to spin the wheel up; the reduced phase voltage available under sinusoidal field-oriented modulation; and the fall of pack voltage through the discharge curve, which must be taken at its lower end if the maneuver is to remain available

on a partly-depleted pack rather than only on a fresh one. The design value is set below the ceiling those effects imply, leaving margin for motor tolerance, bearing and aerodynamic drag, and the residual uncertainty in M . The resulting back-EMF should remain a comfortable fraction of the usable bus voltage, and the electrical frequency $\omega_{\max}p/2\pi$ at p pole pairs must sit inside the driver's commutation capability—both checks to be confirmed once the motor and pack are selected.

4.4 Control Design on the 2D Model

4.4.1 LQR Baseline Control

Linearizing about the upright equilibrium $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$ gives

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x := (\theta_b, \dot{\theta}_b, \dot{\theta}_w), \quad (6)$$

with A, B built from the parameters in Table 5. Zero-order-hold discretization at a rate set by the open-loop unstable pole, with a safety margin, gives $x[k+1] = A_dx[k] + B_d u[k]$. A discrete LQR gain $K_d = (K_{d1}, K_{d2}, K_{d3})$ minimizes

$$J(u) = \sum_k x^T[k]Qx[k] + u^T[k]Ru[k], \quad (7)$$

giving $u[k] = -K_d(\hat{\theta}_b[k], \hat{\theta}_b[k], \hat{\theta}_w[k])$.

The tilt estimate $\hat{\theta}_b$ is obtained from a gyroscope and accelerometer mounted at, or as close as mechanically feasible to, the pivot point rather than at the plate's center. This placement is necessary because an accelerometer at radius r from the pivot measures gravity plus the tangential ($r\ddot{\theta}_b$) and centripetal ($r\dot{\theta}_b^2$) acceleration components induced by the plate's rotation, which corrupt the naive estimate $\tan^{-1}(-a_x/a_y)$ during fast motion (e.g., jump-up or aggressive recovery); placing the sensor at $r \approx 0$ makes both terms negligible. Tilt is estimated with a complementary filter combining gyro integration (drift-prone, unaffected by the terms above) with accelerometer inclination (drift-free, but unreliable during shocks):

$$\hat{\theta}_b[k] = (1 - \gamma)(\hat{\theta}_b[k-1] + \dot{\theta}_{b,\text{gyro}}[k] \Delta t) + \gamma \theta_{b,\text{accel}}[k], \quad (8)$$

where $\theta_{b,\text{accel}}[k] = \tan^{-1}(-a_x[k]/a_y[k])$ is the instantaneous accelerometer-only estimate and $\gamma \in (0, 1)$ is a small weight favoring the gyro term between corrections.

A constant tilt-sensor offset d , including residual bias from imperfect pivot placement or sensor misalignment, would otherwise appear as a nonzero steady-state wheel speed rather than a corrected angle. A low-pass filter state

$$x_f[k+1] = (1 - \alpha)x_f[k] + \alpha\hat{\theta}_b[k] \quad (9)$$

is subtracted from the tilt feedback,

$$u[k] = -K_d(\hat{\theta}_b[k] - x_f[k], \hat{\theta}_b[k], \hat{\theta}_w[k]), \quad (10)$$

so d is absorbed by x_f at steady state, requiring no separate calibration stage. This fixed-gain design is the baseline balancing controller; Table 9 summarizes the control variables.

4.4.2 Gain Scheduling & Mode Transitions

The controller switches from the open-loop jump-up sequence to the LQR law once $|\hat{\theta}_b| < \theta_{\text{th}}$, with hysteresis around θ_{th} to prevent mode chattering. The baseline uses a single fixed gain K_d at the upright equilibrium. If recovery degrades near the edges of the operating range, where the small-angle linearization weakens, gain scheduling will be added: additional gains $K_d^{(i)}$ computed at several linearization points, with the controller interpolating between them based on $\hat{\theta}_b$, reusing the same discrete structure and offset filter.

Table 9: Control-design variables (2D and corner-balancing 3D model).

Symbol	Description
x	State vector
A, B	Continuous-time state/input matrices (linearized model)
A_d, B_d	Discrete-time (ZOH) counterparts of A, B
K_d	LQR feedback gain
Q, R	LQR state/input weighting matrices
$J(u)$	LQR cost function
$\hat{\theta}_b, \hat{\dot{\theta}}_b$	Estimated tilt angle, rate (gyroscope/accelerometer)
$\hat{\dot{\theta}}_w$	Estimated wheel rate (encoder)
a_x, a_y	Raw accelerometer readings (tangential, radial axes)
$\theta_{b,\text{accel}}$	Instantaneous accelerometer-only tilt estimate
$\dot{\theta}_{b,\text{gyro}}$	Raw gyro rate measurement
γ	Complementary-filter weight (accelerometer term)
Δt	Control/estimation loop sample period
x_f, α	Offset-correction filter state, filter coefficient
d	Constant tilt-sensor offset
θ_{th}	Tilt threshold, jump-up $\rightarrow$ balancing switch
$K_d^{(i)}$	Scheduled gain at linearization point i (extension)

4.5 Generalization to the 3D Cube

The dynamics, identification workflow, and control approach validated on the 2D model extend to the full three-dimensional cube, where three orthogonal reaction wheels must stabilize the body about its edge and corner equilibria.

4.5.1 Full-Body Equations of Motion

With $R \in SO(3)$ the cube's orientation and $\omega_b \in \mathbb{R}^3$ its body-frame angular velocity, the attitude kinematics are

$$\dot{R} = R[\omega_b]_{\times}, \quad (11)$$

where $[\cdot]_{\times}$ is the skew-symmetric cross-product matrix. The assembly's angular momentum about the pivot, expressed in the body frame, generalizes the single-axis momentum term of the 2D model:

$$H = I\omega_b + \sum_{i=1}^3 I_{w,i} \omega_{w,i} e_i. \quad (12)$$

Euler's equation gives

$$\dot{H} + \omega_b \times H = \tau_g - C_b \omega_b, \quad (13)$$

and each wheel obeys

$$I_{w,i}(\dot{\omega}_{b,i} + \dot{\omega}_{w,i}) = T_{m,i} - C_{w,i} \omega_{w,i}, \quad i = 1, 2, 3. \quad (14)$$

Table 10 defines all symbols.

Restricting motion to a single vertical plane with two wheel torques set to zero recovers the 2D prototype model exactly, with $I_b, l_b, m_b, m_w, I_w, C_b, C_w$ the corresponding single-axis projections of $I, r_{cm}, m, I_{w,i}, C_{w,i}$.

Table 10: Full-body (3D) model variables.

Symbol	Description
R	Cube orientation relative to world frame
ω_b	Body angular velocity, body frame ($\in \mathbb{R}^3$)
$\omega_{b,i} = e_i^T \omega_b$	Body angular velocity about wheel axis i
$\omega_w = (\omega_{w,1}, \omega_{w,2}, \omega_{w,3})$	Wheel angular velocities relative to body
e_i	Spin axis of wheel i (face normal)
I	3×3 inertia tensor, body+wheels, about pivot
H	Total angular momentum about pivot
$\tau_g = m r_{cm} \times (R^T \hat{g})$	Gravity torque about pivot
m	Total assembly mass
r_{cm}	Body-frame vector, pivot to assembly CoM
$\hat{g}$	World vertical unit vector
C_b	3×3 body damping matrix
$I_{w,i}, C_{w,i}$	Inertia, damping of wheel i
$T_{m,i} = K_{m,i} u_i$	Motor torque, wheel i
$u_i, K_{m,i}$	Motor current, torque constant, wheel i

4.5.2 Edge vs. Corner Equilibria & Linearization

Jump-up proceeds in two stages: Flat-to-Edge, where one wheel brakes, followed by Edge-to-Corner, where a second wheel brakes. **Edge balancing** constrains the body to a single axis, so the dynamics reduce to the 2D form, and the 2D LQR baseline and offset filter apply directly, with m_b, l_b, I_b replaced by their edge equivalents.

Corner balancing is genuinely 3D: all three wheels contribute across three rotational DOF. Linearizing about the corner-up equilibrium $(\theta, \omega_b, \omega_w) = (0, 0, 0)$, with $\theta \in \mathbb{R}^3$ the small orientation error, gives the same state-space form as the 2D case:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x := (\theta, \omega_b, \omega_w) \in \mathbb{R}^9, \quad (15)$$

with $A \in \mathbb{R}^{9 \times 9}$ and $B \in \mathbb{R}^{9 \times 3}$ built from I, r_{cm} , and the per-wheel parameters $I_{w,i}, C_{w,i}, K_{m,i}$ (Table 10). The LQR design generalizes directly: K_d becomes a 3×9 gain matrix, using the same discretization, cost function, and offset filter as Table 9, with Q, R weighting all three tilt axes, body rates, and wheel speeds. The mode-transition logic gains a third mode, Edge-to-Corner jump-up, triggered once edge balancing is stable, followed by a second threshold switch into the corner LQR once estimated orientation nears corner-up.

5 Preliminary Hardware Design

This design is preliminary: final actuator and sensor specifications are not yet confirmed, so the mechanical and electrical architecture is kept modular and largely parametric (Section 1.1).

5.1 System Architecture

Figure 2 shows the closed control loop that ties the four subsystems together. The Teensy 4.1 runs estimation and control at 1 kHz and issues torque commands over CAN-FD to three moteus-n1 drivers, each closing a 30 kHz field-oriented current loop on its MN4006 motor. The resulting reaction torque acts on the cube body; the body’s attitude is sensed by the BMI270 IMU and the rotor angle by the MA600 absolute encoder, closing the loop back to the controller. The two loops run at different rates by design: the fast current loop is delegated entirely to the drivers, so the flight controller need only meet the 1 ms outer-loop deadline of Section 4.4.

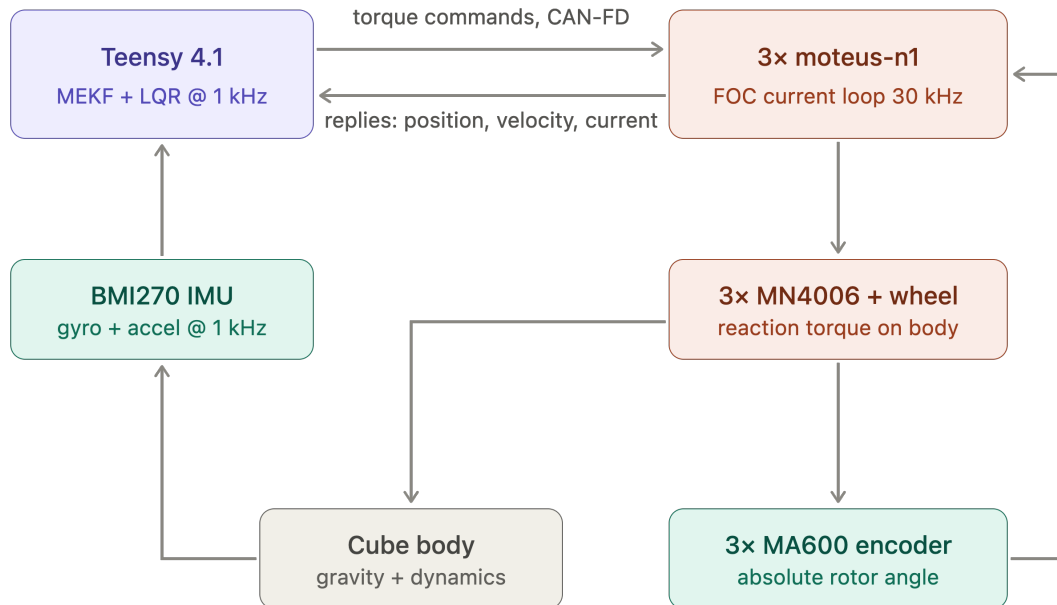


Figure 2: System architecture and control loop. Blue: processing; green: sensing; orange: actuation. Rates shown are the design targets of Section 4.

5.2 Sizing Constraints and Closure Procedure

The cube edge length L and the reaction-wheel inertia I_w are not independent design choices: they are the two ends of a single constraint chain that begins with what must physically be carried inside the cube. This section states that chain and the order in which it is closed. No dimensions are committed here—component selection is still open (Section 5), and the numbers follow from the procedure rather than preceding it.

The battery and the internal structure drive the envelope. The dominant volume claim inside the cube is not the reaction wheels but the power and support hardware. A LiPo pack sized for the jump-up duty cycle is a monolithic, incompressible block: unlike the electronics it cannot be redistributed into the corners or thinned, and its dimensions follow from the required capacity and discharge rating rather than from any packaging preference. Around it must sit an *inner*

structure—a rigid subframe that locates the three orthogonal motors on mutually perpendicular axes through a common centre, carries the three motor drivers, the flight controller, the IMU and the brake servos, and reacts the motor mounting loads into the outer frame. The three wheel/motor modules must not intersect one another, the pack, or this subframe, and the pack should sit near the geometric centre so that it does not skew the cube’s centre of mass away from the body diagonal.

The minimum internal clear volume is therefore set by the pack, the subframe and the electronics stack together, and L is the smallest edge length that contains that volume plus the wall thickness and the corner and edge contact features. This is the first quantity to fix, and everything downstream depends on it.

The envelope caps the wheel, and the wheel must meet the momentum requirement.

Once L is set, the largest wheel outer diameter D_w that clears the subframe, the motor body and the adjacent orthogonal modules is a geometric consequence, and it caps the achievable inertia—not only through $I_w \sim m_w D_w^2/4$, but because any ballast necessarily sits inboard of the ring it displaces, so a wheel that is short of target at its maximum diameter cannot be rescued by adding mass. Against this stands the requirement from the jump-up analysis (Section 4.3.1),

$$I_{w,\text{target}} = \frac{h_w(M, L)}{\omega_{\max}}, \quad h_w \propto ML^{3/2},$$

which grows with both the cube mass and the edge length. Enlarging the cube to gain wheel diameter therefore also raises the requirement—but the achievable inertia grows faster than the requirement does (D_w^2 against $L^{3/2}$), so the loop does converge, and a wheel that is short at a given L can be closed by a modest increase in L .

Mass closes the loop. The cube mass M is not merely a logistics figure: it re-enters the dynamics through the gravitational tip-over torque $\tau_g = MgL/2$, which sets the floor the brake torque must exceed, and through $h_w \propto M$. Growing the envelope to fit a larger wheel adds frame and wheel mass, which raises $I_{w,\text{target}}$ again. Mass, volume and inertia are thus one coupled system, closed by the iteration of Table 11 rather than by any of the three in isolation.

Table 11: Sizing closure procedure. Steps are executed in order; a failure at step 4 returns to step 3 or step 1 as indicated.

Step	Determines	Governing consideration
1	Battery pack geometry	Capacity and discharge rating for the jump-up duty cycle
2	Inner clear volume $V_{\min}$, hence minimum L	Pack + motor subframe + driver/controller stack, non-intersecting; pack near centre
3	Maximum wheel diameter D_w , hence achievable I_w	Clearance to subframe, motor body and the two orthogonal modules
4	Requirement $I_{w,\text{target}} = h_w(M, L)/\omega_{\max}$	Corner jump-up, Eqs. (4)–(5)
5	Accept or iterate	If $I_w < I_{w,\text{target}}$: raise L (step 2), raise $\omega_{\max}$, or trim M ; re-enter at step 3
6	Mass reconciliation	Updated M feeds back into step 4; iterate to a fixed point

Inertia requirement is corner-critical. The two contact cases are not equally demanding. The corner equilibrium requires a larger rise of the centre of mass and a less favourable

momentum-transfer geometry than the edge equilibrium, so for a common maximum wheel speed $\omega_{\max}$ it demands the greater wheel inertia (Table 7). Step 4 above is therefore always evaluated for the corner case; a wheel that satisfies the corner requirement satisfies the edge requirement with margin. Table 12 records both once the procedure closes.

Table 12: Required per-wheel inertia by contact case, to be evaluated at the selected $\omega_{\max}$ once the envelope is fixed.

Contact case	h_w (kg m ² /s)	$I_{w,\text{target}}$ (kg m ²)
Edge	TBD	TBD
Corner (critical)	TBD	TBD

Mass budget. The allocation of Table 13 is the bookkeeping side of steps 5 and 6: it is populated as components are selected and re-totalled on each iteration, since the total is an input to $I_{w,\text{target}}$ and not merely an output of the design.

Table 13: Mass budget (per cube), to be populated as components are selected. The total is an input to the inertia requirement, not only a result.

Subsystem	Allocated mass (g)	Status
Battery pack	TBD	Drives envelope (step 1)
Reaction wheels (3×)	TBD	Pending Sec. 5.3.3
Motors (3×)	TBD	Pending motor selection
Inner structure / motor subframe	TBD	Pending CAD
Outer frame / contact features	TBD	Pending CAD
Electronics (MCU, drivers, IMU, servos)	TBD	Allowance
Wiring / fasteners / margin	TBD	Allowance
Total M	TBD	Closed by iteration

5.3 Mechanical Design

5.3.1 2D Prototype Build

5.3.2 3D Cube Frame

The frame is conceived in two parts, reflecting the constraint chain of Section 5.2. An *inner structure* locates the three motors on mutually perpendicular axes through a common centre and carries the battery pack, the three drivers, the flight controller and the brake servos; it is the element that fixes the minimum internal clear volume and therefore the edge length L . An *outer frame* carries the edge and corner contact features, protects the wheels, and reacts the motor mounting loads from the inner structure into the ground contact. Detailed geometry is pending CAD and follows the envelope closure.

5.3.3 Reaction-Wheel Sizing

The wheel must supply the per-wheel inertia $I_{w,\text{target}}$ demanded by the corner jump-up (Section 4.3.1), at an outer diameter no greater than the envelope permits (step 3 of Table 11), while remaining light enough not to dominate the cube mass—since its own mass feeds back into $I_{w,\text{target}}$ through M . These three demands pull against each other, and the wheel architecture is chosen so that the design can be trimmed against them after fabrication rather than having to be right the first time.

Architecture: ballasted ring. Rather than adjusting inertia by scaling a solid disc, the wheel is built as a printed *ring plus spokes* whose inertia is then tuned by *ballast*: standard steel bolts and nuts seated in blind hexagonal pockets distributed around the ring. Each pocket both removes plastic and adds steel at the largest available radius, so the net inertia gain per pocket is large, and the target is reached by choosing how many bolts to install. This makes wheel inertia a discrete, reversible knob: the physical wheel can be re-tuned in either direction after fabrication to absorb the parameter uncertainty that the sizing procedure necessarily carries—most importantly the residual uncertainty in M , which is not known until the build is complete. A solid disc offers no such adjustment.

The design intent is that the bare printed ring supplies most of the required inertia and the ballast acts only as a trim, so that the bolt count sits well inside its feasible range in both directions.

Parameters to be fixed. The geometry is defined by the quantities of Table 14, all of which follow from the envelope closure of Section 5.2 and are recorded there once fixed.

Table 14: Reaction-wheel design parameters, to be fixed once the envelope closes.

Symbol	Parameter	Value
D_w	Ring outer diameter (capped by envelope)	TBD
w_r	Ring radial width	TBD
t_r	Ring axial thickness	TBD
n_s, w_s	Spoke count and width	TBD
ρ_p	Printed-material density	TBD
r_b	Ballast radius	TBD
m_b	Net mass added per installed bolt	TBD
ΔI_b	Net inertia added per installed bolt	TBD
N	Bolt count reaching $I_{w,\text{target}}$	TBD
m_w	Resulting wheel mass	TBD
I_w	Resulting wheel inertia	TBD

The ballast is characterised *net* of the plastic each pocket displaces: m_b and ΔI_b are the increments the installed bolt-and-nut adds over the material it replaces, evaluated at the ballast radius. Sizing then reduces to solving

$$I_w(D_w, N) = I_{w,\text{ring}}(D_w) + N \Delta I_b \geq I_{w,\text{target}}$$

for the smallest admissible N at each candidate D_w , subject to the fit and mass checks below.

Trade study. Table 15 is the record sheet for the outer-diameter sweep, populated once $I_{w,\text{target}}$ and the envelope cap are known. Each candidate diameter is evaluated on four criteria, and the selected design is the one that passes all of them with the widest trim margin:

1. **Envelope.** D_w must clear the motor subframe, the motor body and the two orthogonal wheel modules inside the selected L .
2. **Inertia.** $I_w \geq I_{w,\text{target}}$, with margin for the uncertainty in M .
3. **Pocket fit.** Enough plastic wall must survive between adjacent pockets and radially inboard and outboard of each. Small diameters fail here: they need so many bolts to reach the target that no wall remains between them.

4. **Mass and trim margin.** The three-wheel set must stay within its allocation in Table 13. Diameters large enough that the bare ring already exceeds the target are rejected—they leave no downward trim authority and cost mass for nothing.

Criteria 3 and 4 bracket the feasible range from below and above respectively, so the selection is the largest diameter that still requires a non-trivial bolt count.

Table 15: Reaction-wheel outer-diameter trade study. To be populated once $I_{w,\text{target}}$ and the envelope cap are fixed; N is the bolt count reaching the target, rounded up to a multiple of the spoke count for balance.

D_w (mm)	N	Mass (g)	I_w (kgm ²)	I_w/target (%)	3 wheels (g)	Pocket fit
TBD	TBD	TBD	TBD	TBD	TBD	TBD
TBD	TBD	TBD	TBD	TBD	TBD	TBD
TBD	TBD	TBD	TBD	TBD	TBD	TBD
TBD	TBD	TBD	TBD	TBD	TBD	TBD
TBD	TBD	TBD	TBD	TBD	TBD	TBD

Pocket placement. The pockets cannot in general be spaced uniformly. A spoke occupies a finite angular width where it meets the ring, as does a pocket at the ballast radius, so a pocket centre must clear a spoke centre by the sum of the two half-widths. Whenever the uniform pitch $360^\circ/N$ divides the spoke pitch exactly, a uniform arrangement places a pocket on every spoke root and is inadmissible. The pockets are therefore distributed in equal groups within each inter-spoke sector, spread over the usable arc that remains after the spoke exclusion zones are removed. Grouping by sector preserves the n_s -fold symmetry and hence static and dynamic balance, and the count per sector is chosen to leave spare positions for later trim.

Bolt retention. Each bolt carries a centrifugal load $F = m_b \omega_{\max}^2 r_b$ at the design speed, reacted by the pocket floor in bending. Positive retention is nonetheless required, and is provided by through-fastening each bolt with its nut against the pocket floor. This both retains the ballast and preserves the removability that motivates the scheme; the floor thickness and the pocket depth are set so that F at the final $\omega_{\max}$ is carried with margin.

5.4 Electrical Design

5.4.1 Components

The electronics and software will be built around a microcontroller communicating over a CAN bus with a motor driver that closes the current loop for the selected brushless motor. Wheel speed will be measured with a magnetic encoder on the motor shaft, and body tilt will be estimated from an onboard IMU. Braking will be actuated by a digital servo driven directly from the microcontroller, and a wireless module will provide a telemetry and debugging link. Power will be supplied by a LiPo battery, stepped down through a DC-DC converter for the logic-level electronics, with the motor driver powered directly from the battery bus.

The full parts list is catalogued in Appendix A (Table 16). The supplied set is an electronics-only BOM; the reaction wheels, the frame and all printed parts are the team’s design responsibility, and their fabrication is the leading schedule risk.

5.4.2 Electrical Diagram

The power architecture is shown in Figure 3. A single 6S LiPo pack (22.2 V nominal, 3600 mA h) feeds everything through one XT90 connector, behind a combined E-stop and fuse using an

XT90-S anti-spark connector to limit inrush into the driver bulk capacitance. The bus then splits into two rails. The *motor bus* runs at pack voltage (22–25 V across the discharge curve) directly into the three moteus drivers, with bulk capacitance rated to at least 50 V to absorb regenerative braking transients—which are not incidental here, since the jump-up maneuver of Section 4.3 dumps the full wheel kinetic energy back into the bus. The *logic rail* is derived by an LM2596 buck converter stepping 22 V down to 5 V for the Teensy, the ESP32 telemetry link, the brake servo and the IMU. Keeping the two rails separate downstream of the E-stop means motor-current ripple does not reach the sensing and processing electronics.

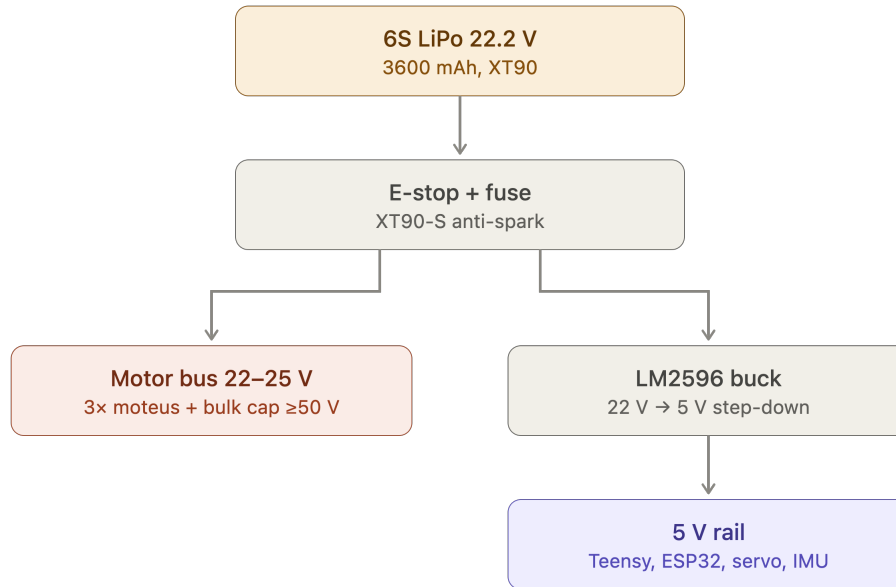


Figure 3: Power distribution tree. A single 6S pack feeds the motor bus directly and the 5 V logic rail through a buck converter, both behind a common E-stop and fuse.

5.5 System Integration

5.5.1 Onboard Software Pipeline

The subsystems of Figure 2 are coordinated by a single deterministic task on the Teensy, whose structure is given in Figure 4. Each 1 ms cycle executes the same fixed sequence: sensor ingest reads the BMI270 over SPI and the moteus replies over CAN; the MEKF propagates attitude on the gyro and corrects it against the accelerometer while estimating gyro bias; the state vector $x = [g_B, \omega, h]$ is assembled from the estimator output and the wheel speeds; the LQR law of Section 4.4 evaluates $u = -Kx$ with a penalty on stored momentum h to bleed off wheel saturation; and the resulting commands leave as CAN torque requests to the drivers and a PWM signal to the brake servo. Telemetry is tapped off the assembled state to SD and Wi-Fi outside the control path, so logging cannot perturb loop timing.

The fixed ordering matters: with no dynamic allocation and no blocking calls, the worst-case cycle time is bounded by construction rather than measured after the fact, which is what makes the 1 kHz sample period of the discrete LQR design a design guarantee rather than an average.

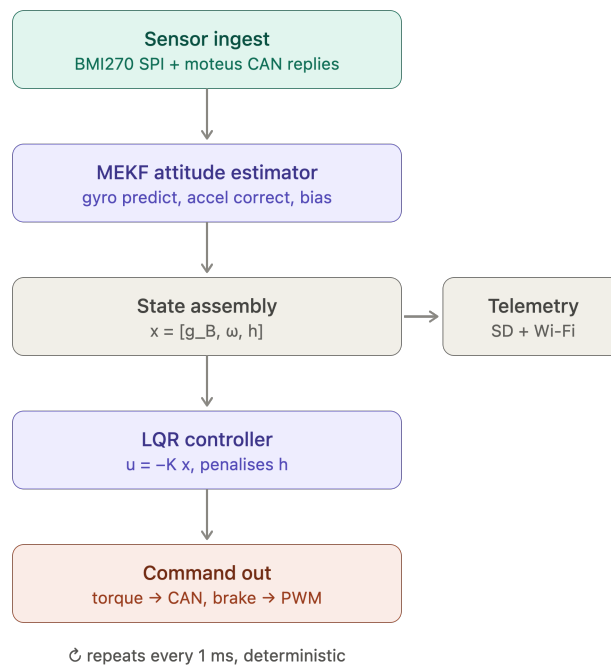


Figure 4: Teensy 4.1 control-loop software pipeline. The estimator and controller stages implement the design of Section 4.4; telemetry is a side-branch off the control path.

6 Testing and Validation Plan

This plan covers system-level validation. Parameter-identification tests that feed the model directly (swing test, current-step identification) are described with the model they validate in Section [4.2](#).

6.1 Simulation-First Validation

6.2 Bring-up Sequence

6.3 Sensor Testing

6.4 Milestone Acceptance Criteria

6.5 Control Performance Evaluation

7 Future Work

7.1 NMPC & Warm-Start Control (Future Implementation)

7.2 Autonomous Stand-up & Disturbance Rejection

7.3 Single-Wheel Variant / Extensions

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A Bill of Materials

The components supplied for this project constitute an *electronics* bill of materials: the actuation chain, wheel and attitude sensing, on-board compute, communications and power distribution are specified, but no structural or mechanical hardware is. The reaction wheels, the frame, the motor mounts, the fasteners and every printed bracket are the team’s design responsibility, and are treated as such in Section 5. Table 16 lists the supplied items grouped by subsystem. The actuation figures derived from these parts — the torque constant, the achievable torque and wheel speed, and the inertia that sizes the wheel — are developed where they are used, in Sections 4.3.1 and 5.3.3.

The architecture the BOM implies is a two-level control stack: each mjbots moteus-n1 closes a high-rate field-oriented current loop on its motor, using the MA600 encoder for rotor angle, while the Teensy 4.1 runs the outer attitude-estimation and state-feedback loop over CAN-FD — the same inner-tachometer / outer-attitude layering used in spacecraft reaction-wheel assemblies. A mechanical wheel brake, following the original ETH Cubli [2], provides the near-impulsive momentum transfer required for the jump-up.

Procurement is the single largest schedule risk. The long-lead items are not electronic but mechanical: the frame stock and the printed structure gate the entire build, and with three actuation sets and no spare, the loss of a motor or driver in service would force the reduced-actuation one-wheel variant [8]. These items are ordered first. Consumables and bench equipment not covered by the supplied set — wire, connectors, fasteners, filament, the current-limited bench supply and the battery-safety kit — are tracked separately in the build documentation.

Table 16: Supplied bill of materials, grouped by subsystem.

Ref	Item	Qty ^a	Key specifications	Function
Actuation				
1	T-Motor MN4006 KV380	3	18N24P outrunner; 380 rpm/V; 68 g; $\varnothing 44.35$ mm $\times$ 21 mm; 194 m Ω ; peak 16 A, 380 W	Reaction-wheel torque source
2	mjbots moteus- n1	3	10–54 V; 100 A peak / 9 A cont.; 15–30 kHz FOC; 5 Mbit/s CAN- FD; 14.6 g	FOC current/torque driver (inner loop)
11	TowerPro MG92B servo	1–3	metal-gear micro servo; ≈ 0.29 N m (≈ 3 kg cm) at 6 V; 50 Hz PWM	Wheel brake actuator (jump-up)
Sensing				
3	mjbots MA600 breakout + ring magnet	3	TMR absolute angle; SPI 6 MHz, 3.3 V; 16-bit (65,536 counts/rev); air-gap ≤ 0.5 mm	Wheel angle: commuta- tion + state feedback
10	SparkFun BMI270 6-DoF IMU	1	3-axis accel + 3-axis gyro (no magnetometer); I ² C 400 kHz (Qwiic); ODR up to ≈ 1 kHz	Body attitude reference (tilt + rates)
Compute and Communications				
4	Teensy 4.1	1	600 MHz Cortex-M7 + FPU; na- tive CAN-FD (CAN3); 5 V supply	Flight computer: esti- mator + control law
5	CAN-FD adapter (Teensy 4.1)	1	CAN-FD transceiver, logic-level $\rightarrow$ differential; 5 Mbit/s	Bus transceiver for the MCU

Continued on next page

Table 16 (continued)

Ref	Item	Qty ^a	Key specifications	Function
6	mjbots mjcanfd-usb- 1x	1	USB ↔ CAN-FD, 5 Mbit/s	Bench interface (cali- bration, tuning)
7	JST-PH3 CAN cables	≥4	TODO: conductor gauge / current rating not in source ; 3-conductor daisy-chain	CAN bus interconnect
8	Seeed XIAO ESP32-C6 + antenna	1	Wi-Fi; 3.3 V UART to MCU; TX burst ≈ 300 mA	Wi-Fi telemetry / com- mand link
9	60.4 Ω 0.5 W resistor	4	split termination, $2 \times 60.4 \Omega \approx$ 120 Ω bus impedance	CAN bus termination
17	4.7 nF capacitor	2	common-mode filter, bus midpoint to ground	CAN split-termination filter
Power				
12	Turnigy 6S LiPo (XT90)	1	22.2 V nom / 25.2 V charged; 3600 mA h; 60C; ≈ 80 W h; ≈ 600 g	Primary energy store
13	LM2596 buck module	1	step-down to 5.0 V; rated 2–3 A (derates without heatsink)	5 V logic rail
14	XT90 connector pair	1–2	rated ≈ 90 A	Battery power connec- tor
15	100 μF 16 V electrolytic	2–4	16 V part — 5 V rail only	5 V-rail bulk reservoir
19	1N5819 Schot- tky diode	1–2	1 A, 40 V	5 V-rail back-feed pro- tection
Integration				
16	100 nF 50 V capacitor	6–10	local decoupling at each IC	Per-IC supply decou- pling
18	Double-sided perfboard	1	15 cm × 9 cm, double-sided	Power / signal distribu- tion board

^a Quantity per assembled cube; motors, drivers and encoders are 3 each. A fourth actuation set (motor + moteus-n1 + MA600) is retained on the 1-DoF bench rig (S1) throughout the build, so four of each are procured.